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AQA GCSE Geography (8035) — Paper 2: Challenges in the Human Environment · 3.2.3 The Challenge of Resource Management (option: energy)

2.7 Energy

1. Areas of Energy Surplus and Deficit

Energy security — having access to reliable, affordable and sufficient sources of energy. **Energy insecurity** — when supply is unreliable, too expensive or insufficient to meet demand. **Energy surplus** — a country produces more energy than it uses (e.g. Saudi Arabia, Russia, Norway, Qatar). **Energy deficit** — a country uses more energy than it produces and must import the rest (e.g. Japan, South Korea, Germany, the UK, India).

- Energy resources are **unevenly distributed**: the Middle East holds around half of the world's proven oil reserves; Russia, Iran and Qatar hold most natural gas; China, the USA, India and Australia have the largest coal reserves.
- Consumption is also uneven: the average person in the USA uses several times more energy than the global average and dozens of times more than someone in Ethiopia or Niger.

2. Reasons for Increasing Energy Consumption

- **Economic development** — as countries industrialise, factories, transport and infrastructure need more energy. China's energy use roughly tripled between 2000 and 2020.
- **Rising population** — more people need more energy for homes and transport.
- **Technology** — more devices (phones, computers, air-conditioning, electric vehicles, data centres) need more electricity.
- **Rising living standards** — wealthier people own more cars and appliances and travel more.

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3. Factors Affecting Energy Supply

Factor	Effect
Physical factors	Fossil fuels are found only where the geology is right; renewable energy depends on geography — e.g. hydro needs high rainfall and steep valleys, geothermal needs tectonic activity, solar needs sunshine. Some reserves are in hostile environments (e.g. the Arctic, deep oceans).
Cost of exploitation and production	Deep-water oil, oil sands and Arctic drilling are expensive; when energy prices fall, these become uneconomic. Renewables have high start-up costs but low running costs.
Technology	New technology opens up previously inaccessible reserves (e.g. fracking for shale gas in the USA) and makes renewables cheaper and more efficient.
Political factors	Conflict and political instability disrupt supply — e.g. Russia's invasion of Ukraine (2022) caused gas prices in Europe to soar. Governments may restrict exploitation for environmental reasons (e.g. the UK fracking moratorium).

4. Impacts of Energy Insecurity

- **Exploitation of environmentally sensitive areas** — e.g. oil drilling in the Arctic (Alaska), extraction of tar sands in Alberta, Canada, which destroys forest and uses huge amounts of water.
- **Economic and environmental costs** — high energy prices raise the cost of living and business costs; countries spend more on imports.
- **Food production** — farming depends on energy for machinery, fertilisers and transport; biofuel crops replace food crops, raising food prices.
- **Industrial output** — power cuts reduce production (e.g. South Africa's "load-shedding" blackouts have damaged its economy).
- **Potential for conflict** — competition over energy resources can cause tension, e.g. disputes in the South China Sea and the Arctic; energy used as a political weapon.

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5. Strategies to Increase Energy Supply

Renewable sources

Source	Advantages	Disadvantages
Wind	No emissions when running; costs have fallen sharply; offshore wind is very productive	Intermittent; visual and noise impact; can harm birds
Solar	Cheap and falling in cost; works at small scale in remote areas	Only generates in daylight; less effective in cloudy places; large solar farms use farmland
Hydroelectric (HEP)	Reliable and controllable; reservoirs can store water	Dams flood valleys, displace people and disrupt river ecosystems
Tidal and wave	Predictable tides; huge potential around the UK	Expensive; can affect estuary habitats; technology still developing
Geothermal	Reliable, low emissions	Only possible in tectonically active areas (e.g. Iceland, Kenya)
Biomass	Uses waste; can be "carbon neutral" if replanted	Crops compete with food for land; burning releases pollution

Non-renewable sources

- **Fossil fuels** (coal, oil, gas) — reliable and flexible, with existing infrastructure, but finite and the largest cause of greenhouse gas emissions and air pollution.
- **Nuclear power** — very low carbon and reliable, but expensive to build, produces radioactive waste and carries a small risk of serious accidents (Chernobyl 1986, Fukushima 2011).

6. Fracking — Advantages and Disadvantages

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Fracking (hydraulic fracturing) — drilling into shale rock and pumping in water, sand and chemicals at high pressure to crack the rock and release natural gas trapped within it.

Advantages	Disadvantages
Could provide a domestic gas supply, reducing imports and improving energy security	Can trigger small earthquakes — operations at Preston New Road, Lancashire, were halted in 2019 after a 2.9 magnitude tremor
Gas produces about half the CO_2 of coal when burned, so it could be a "bridge" fuel	Risk of groundwater contamination from chemicals and methane leaks
Creates jobs and tax revenue; transformed the USA into a major gas exporter	Uses large volumes of water; heavy lorry traffic, noise and visual impact
	Continues dependence on fossil fuels, conflicting with net-zero targets; strong local opposition in the UK

7. Moving Towards a Sustainable Resource Future

Individual energy use and carbon footprints

- A person's **carbon footprint** is the total greenhouse gas emissions caused by their activities — heating, travel, diet and the products they buy.
- Individuals can reduce this by insulating homes, using public transport, cycling or walking, eating less meat and buying fewer new goods.

Energy conservation

Area	Strategies
Designing homes and workplaces	Loft and cavity-wall insulation, double or triple glazing, solar panels, heat pumps, passive solar design (large south-facing windows), green roofs, smart meters.

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	BedZED in London was designed as a zero-fossil-energy development.
Transport	Electric vehicles (new petrol and diesel car sales in the UK are due to end in 2030); integrated public transport; cycle lanes; congestion charges and low-emission zones; car-sharing.
Demand reduction	Switching off appliances; using energy-efficient appliances and LED lighting; smart meters that show energy use in real time.
Use of technology	Improving the efficiency of power stations, vehicles and appliances; carbon capture and storage; smart grids that match supply and demand.

8. Local Renewable Scheme in an LIC: Chambamontera Micro-Hydro Scheme, Peru

Chambamontera, northern Andes, Peru

- A remote, mountainous rural community about 6 hours' drive and walk from the nearest town, where families had no electricity and relied on candles, kerosene and firewood.
- The charity **Practical Action** helped the community build a **micro-hydro scheme**: water from a mountain stream is diverted through a pipe to drive a small turbine, which generates electricity for around 80 homes, a school and health post. The community provided labour; the cost was shared through loans and grants.
- **Benefits:** electric lighting allows children to study and adults to work after dark; fridges keep vaccines and food cold; reduced use of kerosene and firewood improves health (less indoor smoke) and reduces deforestation; small businesses such as milk processing and carpentry have grown.
- **Why it is sustainable:** renewable, run-of-river (no large dam), low-cost to maintain, and locally managed — residents were trained to operate and repair it, and pay a small monthly fee for upkeep.

Exam Tip: For "evaluate" questions on energy strategies, compare large-scale options (nuclear, fracking, offshore wind) with local schemes such as Chambamontera, considering cost, reliability,

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environmental impact and who benefits.

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